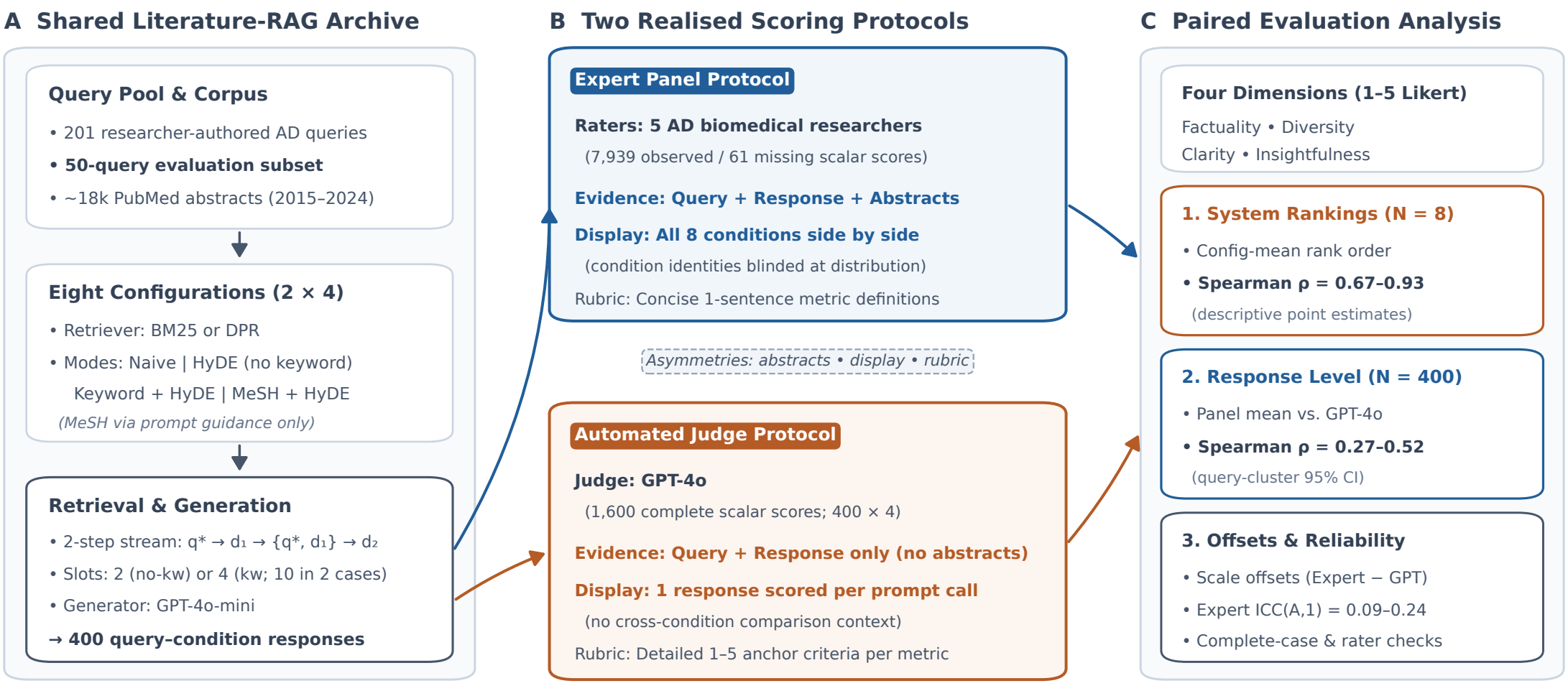
A Shared Literature-RAG Archive
Query Pool & Corpus
• 201 researcher-authored AD queries
• 50-query evaluation subset
• ~18k PubMed abstracts (2015–2024)
Eight Configurations (2 × 4)
• Retriever: BM25 or DPR
• Modes: Naive | HyDE (no keyword)
Keyword + HyDE | MeSH + HyDE
(MeSH via prompt guidance only)
Retrieval & Generation
• 2-step stream: q* → d₁ → {q*, d₁} → d₂
• Slots: 2 (no-kw) or 4 (kw; 10 in 2 cases)
• Generator: GPT-4o-mini
→ 400 query–condition responses
B Two Realised Scoring Protocols
Expert Panel Protocol
Raters: 5 AD biomedical researchers
(7,939 observed / 61 missing scalar scores)
Evidence: Query + Response + Abstracts
Display: All 8 conditions side by side
(condition identities blinded at distribution)
Rubric: Concise 1-sentence metric definitions
Asymmetries: abstracts • display • rubric
Automated Judge Protocol
Judge: GPT-4o
(1,600 complete scalar scores; 400 × 4)
Evidence: Query + Response only (no abstracts)
Display: 1 response scored per prompt call
(no cross-condition comparison context)
Rubric: Detailed 1–5 anchor criteria per metric
C Paired Evaluation Analysis
Four Dimensions (1–5 Likert)
Factuality • Diversity
Clarity • Insightfulness
1. System Rankings (N = 8)
• Config-mean rank order
• Spearman ρ = 0.67–0.93
(descriptive point estimates)
2. Response Level (N = 400)
• Panel mean vs. GPT-4o
• Spearman ρ = 0.27–0.52
(query-cluster 95% CI)
3. Offsets & Reliability
• Scale offsets (Expert − GPT)
• Expert ICC(A,1) = 0.09–0.24
• Complete-case & rater checks